Maulik Madhavi

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Experience

NCS Pte Ltd, Singapore, Senior AI Engineer — Apr 2021 – present, 5 years 3 months

- Contributed to two major platforms under KAI CoCreate: KAI Create (Low-code video analytics training tool) and KAI SnS (Video search and summarization platform:) through computer vision and backend optimization
- Designed and implemented face indexing system supporting million-scale face search capabilities, along with person and vehicle detection models with attribute recognition
- Built prompt studio platform enabling users to efficiently search, summarize, and analyze video content for downstream applications, collaborating across cross-functional teams
- Fine-tuned open-weight LLMs (Mistral-7B, Llama-2-7B) using LoRA-based supervised fine-tuning (SFT) on custom Alpaca-format instruction datasets curated from internal data
- Optimized high-throughput VLM serving with vLLM (PagedAttention, continuous batching), generating descriptions for 48 concurrent 10-second video chunks in ~7 seconds on a single H100 GPU
- Optimized computer-vision recognition and detection models for edge and on-prem deployment using quantization, pruning, and distillation with ONNX Runtime, TensorRT, OpenVINO, and TFLite

National University of Singapore (NUS), Singapore, Research Fellow — Dec 2017 – Apr 2021, 3 years 5 months

- Designed and deployed backend dialogue system for autonomous vehicles supporting multi-modal conversations (FAQ-based, intention-driven, and open-domain) leveraging Google Search API and Wikipedia
- Developed text-dependent speaker verification system optimized for mobile deployment
- Enhanced Automatic Speech Recognition (ASR) system for healthcare applications by adapting language models to Singaporean English, achieving significant Word Error Rate (WER) improvements
- Built automated form-filling system for SCDF call centers using NLU techniques to extract actionable information from ASR-generated transcripts
- Mentored and co-supervised undergraduate and graduate students on research projects

Dhirubhai Ambani Institute of Information and Communication Technology (DA-IICT), Gandhinagar, Project Personnel — Apr 2012 – June 2014, 2 years 3 months

- Prosodically Guided Phonetic Engine Development for Gujarati and Marathi languages (Advisor: Prof. Hemant Patil)
- Developed sophisticated phonetic engine capturing phonetic transcription (IPA), syllabification, and prosodic information (pitch and break marking) across multiple recording modes (read, spontaneous, lecture) from different dialectal regions
- Designed and implemented phonetic engine development workflow using Hidden Markov Models for phoneme recognition, spectral analysis for syllable segmentation, and source characteristic analysis for suprasegmental feature extraction and acoustic modeling
- Built audio search pipeline leveraging the phonetic engine architecture to enable spoken content retrieval and discovery applications for Indian language processing

Education

Ph.D. **Dhirubhai Ambani Institute of Information and Communication Technology (DA-IICT), Gandhinagar**, Information and Communication Technology (ICT) — July 2011 – Nov 2017
- Ph.D. in Information and Communication Technology (ICT)
- CPI: 9.75/10
- Thesis: Design of QbE-STD System : audio representation and matching perspective

M.Tech **Dhirubhai Ambani Institute of Information and Communication Technology (DA-IICT), Gandhinagar**, Information and Communication Technology (ICT) — July 2009 – June 2011
- Master of Technology (M.Tech) in Information and Communication Technology (ICT)
- Communication Systems specialization
- CPI: 8.26/10
- Thesis: Person recognition from their hum

B.E. **V.V.P. Engineering College, Saurashtra University, Rajkot**, Electronics and Communication (EC) — July 2005 – June 2009
- Bachelor of Engineering (B.E.) in Electronics and Communication (EC)
- Percentage: 69.73% (aggregate)

Publications

Vocal Tract Length Normalization using a Gaussian mixture model framework for query-by-example spoken term detection — Nov 2019
M. C. Madhavi, H. A. Patil
10.1016/j.csl.2019.03.005 (Computer, Speech and Language, Elsevier, vol. 58, pp. 175-202)

Design of mixture of GMMs for query-by-example spoken term detection — Nov 2018
M. C. Madhavi, H. A. Patil
10.1016/j.csl.2018.04.006 (Computer, Speech and Language, Elsevier, vol. 52, pp. 41-55)

Combining evidences from magnitude and phase information using VTEO for person recognition using humming — Nov 2018
H. A. Patil, M. C. Madhavi
10.1016/j.csl.2018.04.006 (Computer Speech and Language, Elsevier, vol. 52, pp. 225-256)

Partial matching and search space reduction for QbE-STD — Sept 2017
M. C. Madhavi, H. A. Patil
10.1016/j.csl.2018.04.006 (Computer, Speech & Language, Elsevier, vol. 45, pp. 58-82)

Static and dynamic information derived from source and system features for person recognition from humming — Sept 2012
H. A. Patil, M. C. Madhavi, K. K. Parhi
10.1007/s10772-012-9161-5 (Int. Journal of Speech Technology, Springer, vol. 15, no. 3, pp. 393-406)

Knowledge Distillation from BERT Transformer to Speech Transformer for Intent Classification — Aug-Sept 2021
Y. Jiang, B. Sharma, M. Madhavi, H. Li
10.21437/Interspeech.2021-402 (Interspeech, Brno, Czech Republic, pp. 4713-4717)

Diagnosis of COVID-19 using Auditory Acoustic Cues — Aug-Sept 2021
R. Das, M. Madhavi, H. Li
10.21437/Interspeech.2021-497 (Interspeech, Brno, Czech Republic, pp. 921-925)

Speaker-Utterance Dual Attention for Speaker and Utterance Verification Oct 2020
T. Liu, R. Das, M. Madhavi, S. Shen, H. Li
10.21437/Interspeech.2020-2656 (Interspeech, Shanghai, China, pp. 4293-4297)

Multi-target DoA estimation with an audio-visual fusion mechanism Jun 2021
X. Qian, M. Madhavi, Z. Pan, J. Wang, H. Li
10.1109/ICASSP39728.2021.9413776 (IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Toronto, Canada, pp. 4280-4284)

Leveraging Acoustic and Linguistic Embeddings from Pretrained Speech and Language Models for Intent Classification Jun 2021
B. Sharma, M. Madhavi, H. Li
10.1109/ICASSP46034.2021.9413388 (IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Toronto, Canada, pp. 7498-7502)

Awards

IAPR Travel Scholarship: International Association for Pattern Recognition Travel Scholarship for presenting paper in International Conference on Biometrics, ICB 2012, Delhi, India. Paper: Significance of Magnitude and Phase Information via VTEO for Humming Based Biometrics. (Grant reference: IAPR009)

GATE 2009 Qualification: Qualified in Electronics and Communication Engineering with All India Rank 2142

Invited Talks

Doctoral Consortium Talk: Design of QbE-STD System: Audio Representation and Matching Perspective at Summer School on Speech Signal Processing, DA-IICT, Gandhinagar

Overview of Query-by-Example Spoken Term Detection: Talk at ISCA supported Summer School on Advances in Speech and Audio Processing (ASAP), DA-IICT, Gandhinagar

Query-by-Example Spoken Term Detection - A template matching approach: 5 min Ph.D. competition at ISCA supported summer School on Speech Signal Processing (S4P), DA-IICT, Gandhinagar

Professional Activities

Conference Reviewer: INTERSPEECH (2018-2021), ICASSP 2021, APSIPA 2018, IALP (2019-2020), Computer Speech Language, Speech Communication (Elsevier), IEEE Access

Technologies

Research Areas: Speech signal processing, voice search, speaker verification, spoken language understanding, deep learning for spoken language processing, million-scale indexing

Programming Languages: Bash, C/C++, Python, JavaScript, MATLAB

Deep Learning & ML Frameworks: TensorFlow, PyTorch, OpenCV, scikit-learn, Hugging Face Transformers, NumPy, Pandas, Matplotlib

Large Language Models & AI Tools: LangChain, LlamaIndex, DSPy, vLLM, RASA, n8n

LLM/VLM Fine-tuning & Serving: LoRA/SFT fine-tuning (Mistral-7B, Llama-2-7B), vision-language model tuning (Gemma, moondream, BLIP, CLIP), high-throughput vLLM serving (PagedAttention, continuous batching, KV-cache)

Model Optimization & Edge Deployment: Quantization (INT8/INT4), pruning, knowledge distillation, ONNX Runtime, TensorRT, OpenVINO, TFLite, model export (ONNX/TorchScript)

Web & API Frameworks: Flask, FastAPI, Streamlit, Gradio

Audio/Speech Tools & Systems: Kaldi, wakeup-word detection, voice activity detection

DevOps & Deployment: Git, Docker, Kubernetes

Operating Systems: Windows and Linux